

Lecture 5

Logistic regression

Referred to as Binomial regression in the two class problem.

Goal: Model the probability of being in each class given its predictors by estimating the following functions:

$$P(Y = 1|X = x) = \frac{e^{\beta^T x}}{1 + e^{\beta^T x}}$$

$$P(Y = 0|X = x) = 1 - \frac{e^{\beta^T x}}{1 + e^{\beta^T x}} = \frac{1}{1 + e^{\beta^T x}}$$

Maximum likelihood

Given n data points $\{x_i\}_{i=1}^n$ drawn independently from $p(x; \theta)$, where the form of $p(x)$ is known but θ is unknown, then $\hat{\theta}_{MLE}$ is the Maximum Likelihood Estimate which maximizes the Likelihood of the data.

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \ell(\theta)$$

In this case, we wish to find $\hat{\beta}$ which maximizes $\ell(\beta)$ where

$$\ell(\beta) = \log(L(\beta)) = \sum_{i=1}^n \log(f(x_i; \beta))$$

$$f(x_i; \beta) = \left(\frac{e^{\beta^T x_i}}{1 + e^{\beta^T x_i}} \right)^{y_i} \left(\frac{1}{1 + e^{\beta^T x_i}} \right)^{1 - y_i}$$

Logistic regression

In order to find $\hat{\beta}$ which maximizes $\ell(\beta)$, we set $\frac{\partial \ell}{\partial \beta} = 0$ and solve β .

$$\begin{aligned}\ell(\beta) &= \sum_{i=1}^n \log f(x_i; \beta) \\ &= \sum_{i=1}^n y_i \log \left(\frac{e^{\beta^T x_i}}{1 + e^{\beta^T x_i}} \right) + (1 - y_i) \log \left(\frac{1}{1 + e^{\beta^T x_i}} \right) \\ &= \sum_{i=1}^n y_i \left[\beta^T x_i - \log(1 + e^{\beta^T x_i}) \right] + (1 - y_i) \left[-\log(1 + e^{\beta^T x_i}) \right] \\ &= \sum_{i=1}^n y_i \beta^T x_i - \log(1 + e^{\beta^T x_i})\end{aligned}$$

Logistic regression

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^n y_i x_i^T - \frac{e^{\beta^T x_i}}{1 + e^{\beta^T x_i}} x_i^T$$

We see that $\hat{\beta}$ cannot be found analytically so we can use a numerical method; the Newton Raphson algorithm is widely used:

- 1) Initialize x_0
- 2) $x_{k+1} = x_k - f''(x_k)^{-1} f'(x_k)$
- 3) Repeat until convergence (ie. $|x_{k+1} - x_k| < \epsilon$)

Logistic regression

For convenience, let $p_i = \frac{e^{\beta^T x_i}}{1 + e^{\beta^T x_i}}$ and $1 - p_i = \frac{1}{1 + e^{\beta^T x_i}}$.

Compute the first derivative (Score vector)

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^n (y_i - p_i) x_i^T$$

Compute the second derivative (Hessian matrix)

$$\frac{\partial^2 \ell}{\partial \beta \partial \beta^T} = - \sum_{i=1}^n p_i (1 - p_i) x_i x_i^T$$

Now we can apply the Newton Raphson algorithm to maximize $\ell(\beta)$

$$\beta^{t+1} \leftarrow \beta^t - \left(\frac{\partial^2 \ell}{\partial \beta^t \partial \beta^{tT}} \right)^{-1} \frac{\partial \ell}{\partial \beta^t}$$

Logistic regression

Recalling some matrix algebra, we can convert all summations to matrix operations.

$$\frac{\partial \ell}{\partial \beta} = \mathbf{X}(\mathbf{y} - \mathbf{p})$$

Note: $\mathbf{X}$ is d by n .

$$\frac{\partial^2 \ell}{\partial \beta \partial \beta^T} = -\mathbf{XWX}^T; W_{ii} = p_i(1 - p_i), W_{ij} = 0$$

The Newton-Raphson algorithm can now be expressed as

$$\beta^{t+1} \leftarrow \beta^t + (\mathbf{XWX}^T)^{-1} \mathbf{X}(\mathbf{y} - \mathbf{p})$$

Logistic regression

Alternatively, the algorithm can be expressed as:

$$\begin{aligned}\beta^{t+1} &\leftarrow \beta^t + (\mathbf{X}\mathbf{W}\mathbf{X}^T)^{-1}\mathbf{X}(\mathbf{y} - \mathbf{p}) \\ &\leftarrow (\mathbf{X}\mathbf{W}\mathbf{X}^T)^{-1}[\mathbf{X}\mathbf{W}\mathbf{X}^T\beta^t + \mathbf{X}(\mathbf{y} - \mathbf{p})] \\ &\leftarrow (\mathbf{X}\mathbf{W}\mathbf{X}^T)^{-1}\mathbf{X}\mathbf{W}\mathbf{Z}\end{aligned}$$

where $\mathbf{Z} = \mathbf{X}^T\beta^t + \mathbf{W}^{-1}(\mathbf{y} - \mathbf{p})$

This algorithm is also known as Iteratively Re-weighted Least Squares (IRLS)

$$\beta^{new} \leftarrow \arg \min_{\beta} (\mathbf{Z} - \mathbf{X}\beta)^T \mathbf{W} (\mathbf{Z} - \mathbf{X}\beta)$$

Logistic regression

Note: For a d -dimensional $\mathbf{x}$ this model has d adjustable parameters. By contrast to LDA we have: $2d$ parameters for the means and $d(d+1)/2$ parameters for the covariance matrix. Together with the class priors, LDA gives a total of $d(d+5)/2 + 1$ parameters which grows quadratically in d , in contrast to the linear growth of parameters (d parameters) of logistic regression. For large d , there is a clear advantage for working with the logistic regression model directly.